

BUCAD 项目报告汇总

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报告目录: C:\Users\876762330\Desktop\Agent\artifacts\reports

说明

本文件汇总项目主要验证、评估、可视化、发布和答辩相关报告，便于阅读、归档和展示。

- 纳入文件: final_validation.md, report_tables.md, threshold_analysis.md, visual_evidence_review.md, release_v1_manifest.md, demo_rehearsal.md, comparison_results.json, busi_eval.json, train_cls_fold1.json, train_seg_fold1.json, busbra_split_summary.json

源报告: final_validation.md

Final Validation Status

Date: 2026-04-23

Environment

- Validation executed in Conda environment BUCAD
- Python 3.11.15
- OS Windows 10
- CUDA available: True
- GPU: NVIDIA GeForce RTX 5060 Laptop GPU

Note: the plan and quickstart were originally written for Python 3.10, but this final validation was completed on Python 3.11.15 per the current local environment decision on 2026-04-23.

Quickstart Validation

The following commands were executed successfully in the current environment:

- python check_env.py
- python check_all.py
- python scripts\make_split.py --config configs\paths.local.yml
- python scripts\train_cls.py --config configs\classifier\baseline.yml --fold 1 --epochs 1
- python scripts\train_seg.py --config configs\segmenter\unet.yml --fold 1 --epochs 1
- python scripts\eval_busi.py --config configs\inference\demo.yml
- Source Gradio smoke: build_app().launch(prevent_thread_lock=True, server_name='127.0.0.1', server_port=7861, share=False) then closed cleanly
- pyinstaller packaging\demo.spec --noconfirm
- python scripts\export_demo_assets.py --config configs\paths.local.yml --output-dir dist\bucad-demo
- Packaged demo smoke: dist\bucad-demo\bucad-demo.exe stayed alive for 20 seconds with no stderr after exporting runtime configs and checkpoints into dist\bucad-demo

Automated Regression Status

- Full test suite result after report document export work: 32 passed in 9.11s

Metrics And Outputs

- BUSBRA split generation:
- samples: 1875
- unique cases: 1064
- folds: 5
- leakage detected: false
- Classifier fold 1 smoke training:
- checkpoint: artifacts/checkpoints/classifier_fold1.pt
- validation AUC: 0.8062
- validation sensitivity: 0.4098
- validation specificity: 0.9051
- validation accuracy: 0.7440
- Segmenter fold 1 smoke training:
- checkpoint: artifacts/checkpoints/segmenter_fold1.pt
- validation Dice: 0.8100
- BUSI external evaluation:
- report: artifacts/reports/busi_eval.json
- sample count: 647
- AUC: 0.7564
- sensitivity: 0.6095
- specificity: 0.8009
- accuracy: 0.7388
- Single-image demo evidence:
- input: BUSI/malignant/malignant (1).png from the local evaluation dataset root
- report: artifacts/reports/demo_single_image/diagnosis.json
- generated visuals:
- artifacts/reports/demo_single_image/original.png
- artifacts/reports/demo_single_image/lesion_overlay.png
- artifacts/reports/demo_single_image/explanation.png

Observations

- The end-to-end system now completes classification, lesion overlay generation, explanation overlay generation, Gradio launch, and packaged demo launch in the validated environment.
- The real single-image demo run produced a complete response with both visualization outputs, but the sampled malignant BUSI image malignant (1).png was classified as benign with malignant probability 0.3769. This is acceptable for smoke validation, but it reinforces that the prototype is not clinically reliable and must remain an auxiliary demo system only.
- Packaging required runtime asset export into dist\bucad-demo so the executable could resolve configs/inference/demo.yml and the trained checkpoints from the packaged working directory. The quickstart and README were updated to match this validated flow.

Handbook Follow-up Validation

Date: 2026-04-23

- Added handbook follow-up implementation paths for EfficientNetV2-S, comparison experiments, threshold analysis, visual evidence export, release manifests, and batch inference.
- Automated regression after handbook follow-up: 30 passed in 9.28s.
- Handbook smoke commands completed:
 - `python scripts\run_comparison.py --config configs\classifier\comparison.yml --fold 1 --model-limit 1 --dry-run`
 - `python scripts\export_demo_assets.py --config configs\paths.local.yml --output-dir artifacts\release_v1`
 - `python scripts\export_visual_evidence.py --config configs\inference\demo.yml --output-dir artifacts\reports\visual_evidence --limit 1`
 - `python scripts\eval_busi.py --config configs\inference\demo.yml`
- BUSI threshold analysis generated artifacts/reports/threshold_analysis.md with best Youden threshold 0.50.
- Full model-selection claims still require the long comparison run with `--epochs 20` or the final agreed epoch budget before report submission.

Report Document Export Validation

Date: 2026-04-23

- Added readable DOCX/PDF report export through `scripts\export_report_documents.py`.
- Generated artifacts\reports\documents\bucad_report_summary.docx for editable report review.
- Generated artifacts\reports\documents\bucad_report_summary.pdf for direct reading, handoff, and defense preparation.
- Rendered PDF pages with PyMuPDF for visual spot-checking because pdftoppm and soffice were not available on this Windows environment.
- Automated validation completed with `python -m pytest tests\integration\test_report_documents.py`: 2 passed.
- Full regression completed with `python -m pytest tests\unit tests\smoke tests\integration`: 32 passed in 9.11s.

源报告: report_tables.md

Report Tables

Status: BUSI threshold table populated from the current validated checkpoint.

Full handbook comparison still requires the long GPU run.

Model Comparison Table

The current smoke evidence validates the comparison pipeline with `--dry-run`.

Populate metric columns after the full handbook comparison run.

Model	Fold	AUC	Sensitivity	Specificity	Runtime Seconds
Basic CNN	1	pending	pending	pending	0.003 dry-run

BUSI Evaluation Table

Generated from artifacts/reports/busi_eval.json and

artifacts/reports/threshold_analysis.md.

Threshold	AUC	Sensitivity	Specificity	Accuracy
0.50	0.7564	0.6095	0.8009	0.7388

源报告：threshold_analysis.md

Threshold Analysis

This report is generated from BUSI external evaluation outputs.

Default Threshold

- Threshold: 0.50
- AUC: 0.7564291162689332
- Sensitivity: 0.6095
- Specificity: 0.8009

Best Threshold By Youden J

- Threshold: 0.50
- Youden J: 0.4104
- Sensitivity: 0.6095
- Specificity: 0.8009

Sweep

Threshold	Sensitivity	Specificity	Accuracy	Youden J
0.10	1.0000	0.0092	0.3308	0.0092
0.15	1.0000	0.0229	0.3400	0.0229
0.20	0.9905	0.0503	0.3555	0.0408
0.25	0.9810	0.0870	0.3771	0.0679
0.30	0.9810	0.1556	0.4235	0.1366
0.35	0.9286	0.2471	0.4683	0.1757
0.40	0.8571	0.3661	0.5255	0.2233
0.45	0.7333	0.6133	0.6522	0.3466
0.50	0.6095	0.8009	0.7388	0.4104
0.55	0.3143	0.9497	0.7434	0.2639
0.60	0.0571	0.9931	0.6893	0.0503
0.65	0.0048	1.0000	0.6770	0.0048
0.70	0.0000	1.0000	0.6754	0.0000
0.75	0.0000	1.0000	0.6754	0.0000
0.80	0.0000	1.0000	0.6754	0.0000
0.85	0.0000	1.0000	0.6754	0.0000
0.90	0.0000	1.0000	0.6754	0.0000

源报告： visual_evidence_review.md

Visual Evidence Review

Output directory: C:\Users\876762330\Desktop\Agent\artifacts\reports\visual_evidence

Sample	Status	Label	Lesion Missing Reason	Explanation Missing Reason
benign (1)	completed	benign		

源报告： release_v1_manifest.md

Release v1 Manifest

Release directory: C:\Users\876762330\Desktop\Agent\artifacts\release_v1

File	Size Bytes	SHA-256
artifacts/checkpoints/classifier_fold1.pt	44791947	83947356a461d1f903516d3aee7c07b1d2ace8147008f12d660de9bd5394cc5e
artifacts/checkpoints/segmenter_fold1.pt	57426099	36d018d628c587e797f62c3263ef08833de3220879b061d699bbc755aed689dc
configs/classifier/baseline.yml	481	96bed434b616e00941eae0c976c18f9894a9504c1bade242110a3a7e13e45b58
configs/classifier/comparison.yml	1200	969a7d0f3e5625475c9ba19d86dc1625b27607b4e1b831af9ce0decf75e6835c
configs/classifier/efficientnetv2_s.yml	614	0b8dbeea3c1014adb930dbe4424a09e75aad80acdb32e7618d1a9db5d44f8ffb
configs/inference/demo.yml	440	adf137571577664dbdb295f30ada74f7facd633efcb2bc1d6665733909778995
configs/paths.example.yml	382	b27c65a4d5aab47e6c9db863e57f25ac48aead321925d00bb725e7016c8a41c9
configs/paths.local.yml	382	b27c65a4d5aab47e6c9db863e57f25ac48aead321925d00bb725e7016c8a41c9
configs/segmenter/unet.yml	487	071508dde3ac86ffd42ceca39df319c349a00bd8972dc6ff8e66d5f1af34ac22

源报告： demo_rehearsal.md

Demo Rehearsal Checklist

Status: smoke evidence recorded; final live rehearsal with frozen checkpoints still recommended before presentation.

- [x] Confirm automated Gradio smoke test passes.
- [x] Export one benign BUSI visual evidence sample.
- [] Upload one malignant BUSI image and save the result screenshot.
- [x] Confirm lesion overlay is present or the missing reason is visible.
- [x] Confirm Grad-CAM heatmap is present or the missing reason is visible.
- [x] Confirm packaged assets exist under artifacts\release_v1\.
- [x] Confirm release_v1.sha256 is available for handoff verification.

- [] Confirm the auxiliary-use disclaimer is visible in a live browser rehearsal.

结构化报告： comparison_results.json

训练/验证折： 1

摘要字段

字段	值
dry_run	True
model_count	1

模型对比结果

模型	状态	AUC	Sensitivity	Specificity	运行时间(秒)
basic_cnn	validated				0.0050

结构化报告： busi_eval.json

样本数量： 647

核心指标

指标	数值
auc	0.7564
threshold	0.5000
sensitivity	0.6095
specificity	0.8009
accuracy	0.7388
confusion.tn	350
confusion.fp	87
confusion.fn	82
confusion.tp	128

Youden J 最优阈值

指标	数值
auc	0.7564
threshold	0.5000
sensitivity	0.6095
specificity	0.8009
accuracy	0.7388
youden_j	0.4104

指标	数值
confusion.tn	350
confusion.fp	87
confusion.fn	82
confusion.tp	128

结构化报告： train_cls_fold1.json

训练/验证折： 1

摘要字段

字段	值
train_size	1500
val_size	375
device	cuda
checkpoint_path	C:\Users\876762330\Desktop\Agent\artifacts\checkpoints\classifier_fold1.pt

核心指标

指标	数值
auc	0.8062
threshold	0.5000
sensitivity	0.4098
specificity	0.9051
accuracy	0.7440
confusion.tn	229
confusion.fp	24
confusion.fn	72
confusion.tp	50

结构化报告： train_seg_fold1.json

训练/验证折： 1

摘要字段

字段	值
train_size	1500
val_size	375
device	cuda
checkpoint_path	C:\Users\876762330\Desktop\Agent\artifacts\checkpoints\segmenter_fold1.pt

核心指标

指标	数值
dice	0.8100

结构化报告： busbra_split_summary.json

样本数量： 1875

摘要字段

字段	值
fold_count	5
unique_cases	1064
leakage_detected	False
output_path	C:\Users\876762330\Desktop\Agent\artifacts\reports\busbra_5fold_splits.csv